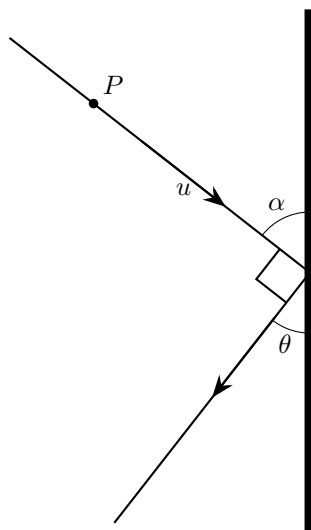


Instructions

Unless a question states otherwise, take $g = 9.8 \text{ m s}^{-2}$ wherever a numerical value of g is required.
Solutions also give final answers for $g = 10 \text{ m s}^{-2}$ where these differ.

Questions

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1. A particle P of mass m is moving with speed u on a fixed smooth horizontal surface. It collides with a fixed smooth vertical barrier. Before impact its path makes an angle α with the barrier, and after impact its path makes an angle θ with the barrier. The incident path and the outgoing path are perpendicular. The coefficient of restitution between P and the barrier is e . The particle loses 36% of its kinetic energy in the collision.

Find the value of e and the exact value of $\tan \alpha$.

[5]

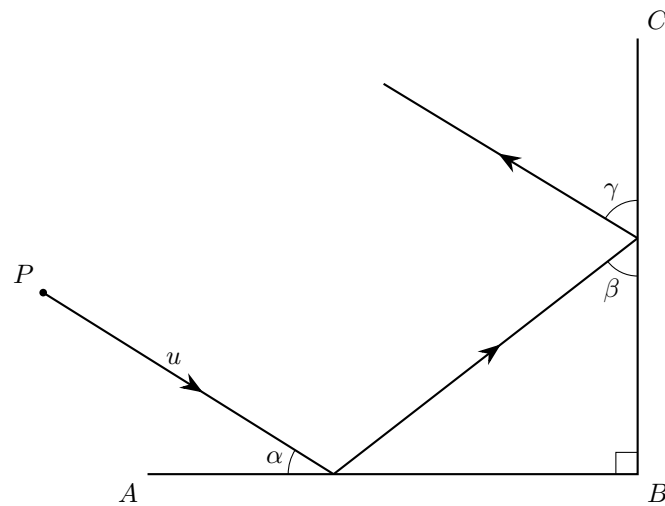
- 2.** A hockey puck moving on a smooth horizontal rink collides with a fixed straight side-board.
Let $\mathbf{i}$ be a unit vector parallel to the side-board and let $\mathbf{j}$ be a unit vector perpendicular to the side-board, directed away from it.

Immediately before the collision, the puck has velocity $(9\mathbf{i} - 12\mathbf{j}) \text{ m s}^{-1}$.

Immediately after the collision, the puck moves with speed 10 m s^{-1} in the direction of the vector $3\mathbf{i} + 4\mathbf{j}$.

Model the puck as a particle.

- (a) Find the coefficient of restitution between the puck and the side-board. [3]
- (b) Determine whether or not the side-board is smooth. [3]
Fully justify your answer.



3. The smooth walls AB and BC are at right angles to each other. A particle P moves on a smooth horizontal floor with speed u and strikes the wall AB at an angle α to AB . It rebounds, then strikes the wall BC at an angle β to BC . After the second impact it rebounds at an angle γ to BC , as shown in the diagram. The coefficient of restitution between P and each wall is e .

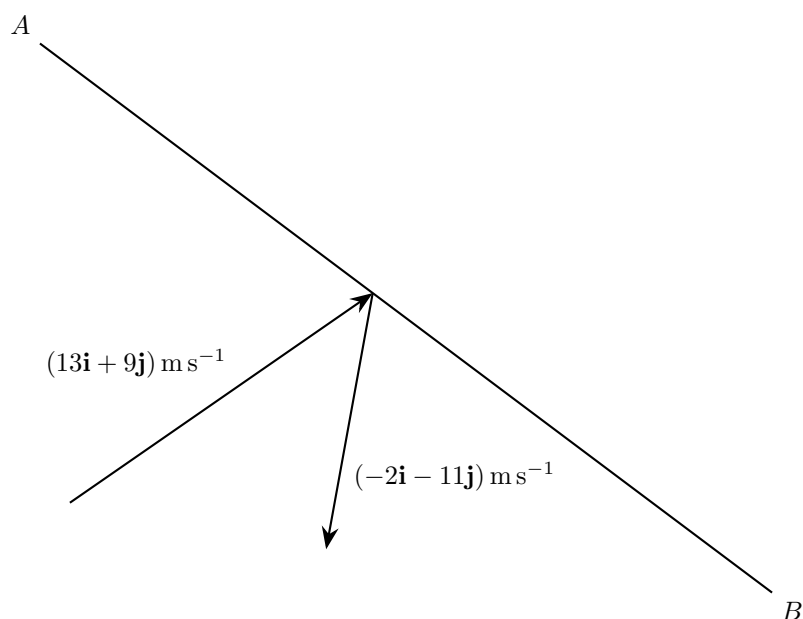
(a) Show that

$$\tan \alpha \tan \beta = \frac{1}{e} \quad [3]$$

(b) Express γ in terms of α and explain what this implies about the final direction of motion of P . [4]

During the first impact the particle loses four times as much kinetic energy as it loses during the second impact. After the second impact the kinetic energy of P is $\frac{9}{25}$ of its initial kinetic energy.

(c) Find the value of e and the value of $\tan \alpha$. [4]



4. The diagram above shows the plan view of part of a smooth horizontal table. The line segment AB represents a fixed smooth vertical wall.

A small ball of mass 0.4 kg moves across the table and rebounds after striking the wall.

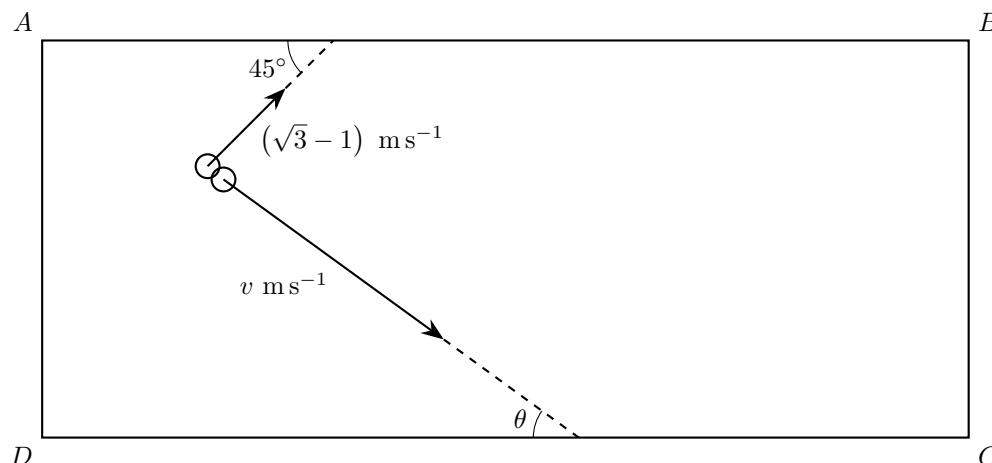
Immediately before impact its velocity is $(13\mathbf{i} + 9\mathbf{j}) \text{ m s}^{-1}$.

Immediately after impact its velocity is $(-2\mathbf{i} - 11\mathbf{j}) \text{ m s}^{-1}$.

The coefficient of restitution between the ball and the wall is e .

(a) Show that AB is parallel to $(4\mathbf{i} - 3\mathbf{j})$. [4]

(b) Find the value of e . [5]



5. Two snooker balls, one white and one red, have equal mass.
The balls are on a horizontal table $ABCD$.

The white ball is struck so that it moves at a speed of 2 m s^{-1} in a direction making an angle of 15° with AB , towards CD .

The white ball hits a stationary red ball.

After the collision, the white ball moves at a speed of $(\sqrt{3} - 1) \text{ m s}^{-1}$ and at an angle of 45° to AB .

After the collision, the red ball moves at a speed $v \text{ m s}^{-1}$ and at an angle θ to CD .

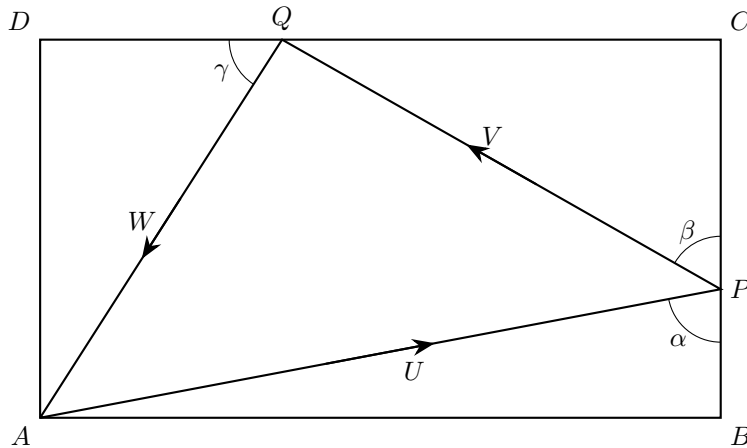
When the collision takes place, the point of impact is twice as far from CD as it is from AB .

The diagram above shows the velocities of the balls after the collision.

After the collision, the white ball hits AB and the red ball hits CD .

Model the balls as particles and neglect any resistance.

- (a) Explain why the two balls hit the sides of the table at the same time. [2]
- (b) Show that $\theta = 36.2^\circ$ correct to one decimal place. [4]
- (c) Find v . [2]
- (d) Determine which ball travels the greater distance after the collision and before hitting the side of the table.
Fully justify your answer. [2]
- (e) State one possible refinement to the model that you have used. [1]



6. A small smooth billiard ball is projected from the corner A of a horizontal rectangular table $ABCD$.

The ball first hits the side BC at the point P , then hits the side CD at the point Q , and then returns to A .

Angle $APB = \alpha$, angle $CPQ = \beta$ and angle $DQA = \gamma$.

The ball moves along AP with speed U , along PQ with speed V and along QA with speed W , as shown in the diagram.

The coefficient of restitution between the ball and side BC is e_1 .

The coefficient of restitution between the ball and side CD is e_2 .

The ball is modelled as a particle.

(a) Show that $\tan \beta = e_1 \tan \alpha$. [4]

(b) Hence show that $e_1 \tan \alpha \tan \gamma = e_2$. [3]

(c) By considering $\angle PAQ$ or otherwise, show that it is possible for the ball to return to A only if $e_2 > e_1$. [6]

If instead $e_1 = e_2$, the ball would not return to A .

(d) Given that $e_1 = e_2$, use the result from part (b) to describe the path of the ball after it hits CD at Q , explaining your answer. [1]

7. A particle P is moving with velocity $(7\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$ on a smooth horizontal plane. It collides with a smooth vertical wall which is parallel to the direction vector $2\mathbf{i} + \mathbf{j}$ and rebounds with velocity $(2\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$.

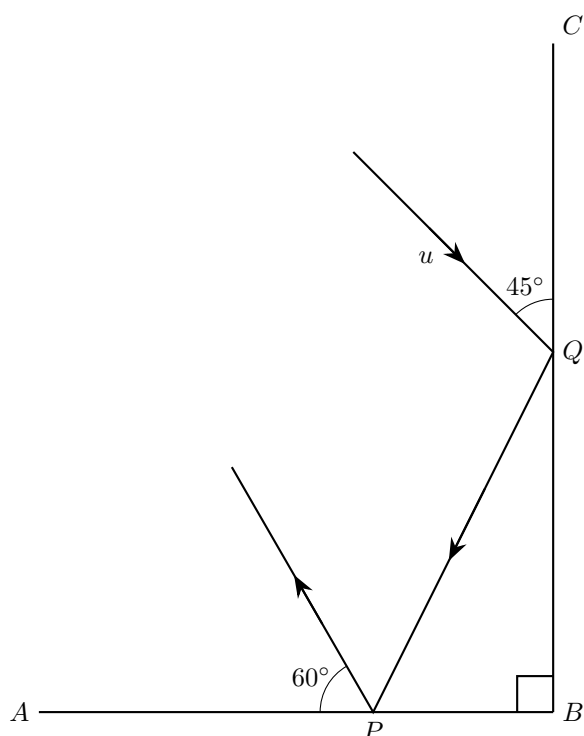
The coefficient of restitution between P and this wall is e .

- (a) Find the value of e . [5]

After this collision, P goes on to hit a second smooth vertical wall which is parallel to the direction vector $\mathbf{i} - 2\mathbf{j}$. The coefficient of restitution between P and this second wall is $\frac{1}{2}$.

The angle through which the direction of motion of P is deflected by its collision with this second wall is α° .

- (b) Find the value of α , giving your answer to the nearest whole number. [5]



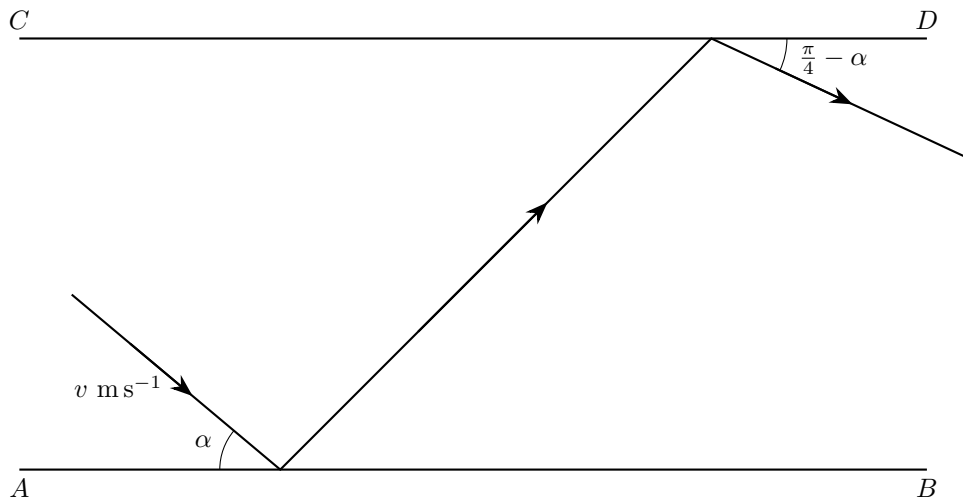
8. The diagram above represents the plan view of part of a horizontal floor, where AB and BC are perpendicular vertical walls.
The floor and the walls are modelled as smooth.

A ball is projected along the floor towards BC with speed $u \text{ m s}^{-1}$ on a path at an angle of 45° to BC . The ball hits BC at Q and then hits AB at P . After striking AB , the ball moves away on a path at an angle of 60° to AB .

The ball is modelled as a particle.

The coefficient of restitution between the ball and wall BC is $\frac{1}{2}$.

- (a) Find the coefficient of restitution between the ball and wall AB and hence show that, using this model, the final kinetic energy of the ball is 50% of the initial kinetic energy of the ball. [8]
- (b) In practice the walls and the floor may not be smooth. Explain how this model is likely to affect the percentage of kinetic energy remaining. [1]



9. The diagram above represents the plan view of part of a horizontal floor, where AB and CD represent fixed vertical walls, with AB parallel to CD .

A small ball is projected along the floor towards wall AB . Immediately before hitting wall AB , the ball is moving with speed $v \text{ m s}^{-1}$ at an angle α to AB , where $0 < \alpha < \frac{\pi}{4}$.

The ball hits wall AB and then hits wall CD .

Immediately after the impact with wall CD , the ball is moving at an angle $\frac{\pi}{4} - \alpha$ to CD .

The coefficient of restitution between the ball and wall AB is $\frac{3}{5}$.

The coefficient of restitution between the ball and wall CD is $\frac{1}{2}$.

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

- (a) Show that

$$\tan \alpha = \frac{2}{3} \quad [7]$$

- (b) Find the percentage of the ball's initial kinetic energy that is lost in the two impacts. [4]

- 10.** A particle P of mass m is moving downwards and to the right in a direction making an angle 45° with the horizontal when it strikes a fixed smooth inclined plane. The plane is inclined to the horizontal at an angle α , where $0 < \alpha < 45^\circ$.

At the instant immediately before the impact, the speed of P is u .

At the instant immediately after the impact, P is moving upwards and to the right in a direction making an angle 45° with the horizontal, with speed v .

- (a) Show that the magnitude of the impulse exerted on the plane by P is $mu \sec(45^\circ - \alpha)$. [5]

The coefficient of restitution between P and the plane is e , where $0 < e < 1$.

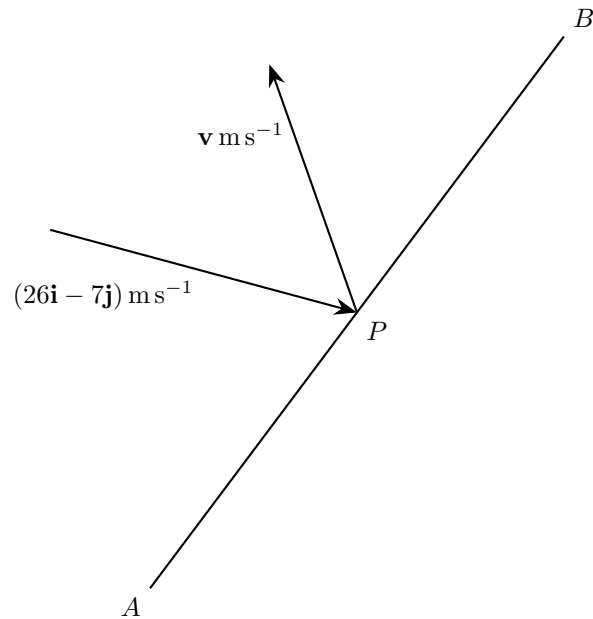
- (b) Show that [3]

$$v^2 = u^2 (\cos^2(45^\circ + \alpha) + e^2 \sin^2(45^\circ + \alpha))$$

- (c) Show that the kinetic energy lost by P in the impact is

$$\frac{1}{2}mu^2(1 - e^2) \sin^2(45^\circ + \alpha) \quad [2]$$

- (d) Hence find, in terms of m , u and e only, the kinetic energy lost by P in the impact. [2]



11. The diagram above represents the plan view of part of a smooth horizontal billiards table, where AB is a fixed smooth cushion.

The direction of $\overrightarrow{AB}$ is parallel to the vector $(3\mathbf{i} + 4\mathbf{j})$.

A small disc of mass 0.2 kg is moving on the table when it strikes the cushion AB .

Immediately before impact with AB , the velocity of the disc is $(26\mathbf{i} - 7\mathbf{j}) \text{ m s}^{-1}$.

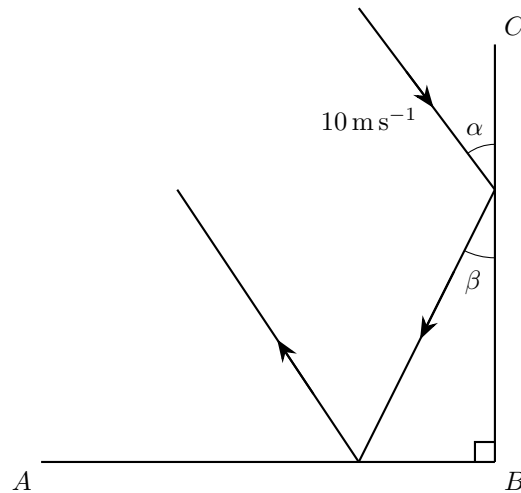
Immediately after impact with AB , the velocity of the disc is $\mathbf{v} \text{ m s}^{-1}$.

The coefficient of restitution between the disc and the cushion is $\frac{3}{5}$.

By modelling the disc as a particle,

(a) show that $\mathbf{v} = -6\mathbf{i} + 17\mathbf{j}$ [6]

(b) Find the magnitude of the impulse exerted by the cushion on the disc in the impact. [3]



- 12.** The diagram above represents the plan view of one corner of a smooth air-hockey table, where AB and BC are fixed cushions with AB perpendicular to BC .

A small puck is struck along the table towards BC with speed 10 m s^{-1} on a path that makes an angle α with BC , where $\tan \alpha = \frac{3}{4}$. The puck hits BC and then hits AB .

Immediately after hitting BC , the puck is moving at an angle β to BC , where $\tan \beta = \frac{1}{2}$.

The coefficient of restitution between the puck and BC is e .

The coefficient of restitution between the puck and AB is $\frac{3}{4}$.

By modelling the puck as a particle and the table and cushions as being smooth,

- (a) show that $e = \frac{2}{3}$ [5]
- (b) find the speed of the puck immediately after it hits AB . [4]
- (c) Suggest two ways in which the model could be refined to make it more realistic. [2]